

Computer Encoding Scheme

Week 3

To make a computer work,
information needs to be converted
into a format that can be understood
by the computer.

Topics

1. How does a computer process data?
2. Binary Digits: 0, 1
3. What is bit, byte, words?
4. What is 32-bit or 64-bit machines, what are the representations of the elements?
5. ASCII Code, Extended ASCII Characters
6. UTF-8 and Unicode
7. Programming Languages & Binary Codes
8. Representing a program in Binary
9. Why study the different number systems (binary, octal, hex)?

Objectives

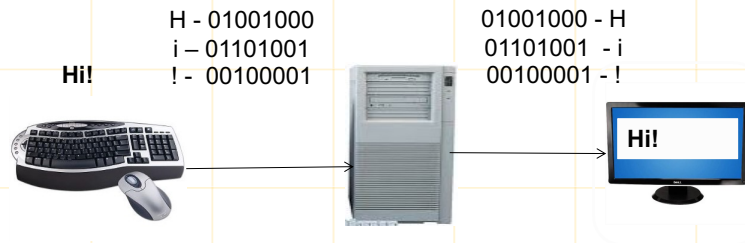
At the end of this lecture, you should have

- A good understanding of how programming languages, codes, numbers, characters are represented in the computer system
- Relationship between programming languages and binary numbers.
- Understand computer's basic data units such as binary numbers, bits, bytes, words

Introduction

How does a computer process data?

All letters/characters/symbols that we type in our computer are translated into a language that computers can understand, which is a machine language - a stream of binary numbers 0 or 1.



Machine language is the lowest-level programming language and the only language understood by computers.

Binary Digits (Bits), Bytes, and Words

Bit (means Binary Digit)

- A "bit" is the smallest unit of digital information in computing.
- Bits are often used in binary logic, representing the basic building blocks of digital information.
- ✓ Bit is used to represent the two levels of status in computer's circuits:
 - 0 – represents 0 V or "OFF" or "false"
 - 1 – represents 5 V or "ON" or "true"
- ✓ 2 bits can represent 4 different values
 - 00, 01, 10, 11

Byte (means "by eight" bits)

- A "byte" consists of 8 bits and can represent 256 characters (0000 0000 to 1111 1111).
- It is a common unit for representing data (e.g. text, integers, etc) in most computer systems .

Binary Digits

1 bit = 2 values: 0, 1

2 bits = 4 values: 00, 01, 10, 11

3 bits = 8 values: 000, 001, 010, 011,
100, 101, 110, 111

...

8 bits = 255 values: 00000000

00000001

00000010

00000011

...

....

11111111

Binary Digits (Bits), Bytes, and Words

Word:

- A "word" is a unit of data that represents a natural size for data processing by a computer's CPU (Central Processing Unit)
- It is the amount of data that a CPU can process per operation (e.g., addition, subtraction, or comparison).
- Word sizes = 16, 32, or 64 bits depending on computer architectures.
- Large word sizes computers → larger numbers & more complex data

Size (bit)	Values	Min	Max
8	255	-128	127
16	65,535	-32,768	32,767
32	4.29E+09	-2.15E+09	2.15E+09
64	1.84E+19	-9.22E+18	9.22E+18

- -128 = 1000 0000
- 127 = 0111 1111
- 32,767 = 0111 1111 1111 1111
- -32,768 = 1000 0000 0000 0000

Binary Digits

The two-digit binary numbers is the basis of machine representation of:

1. Numbers: 0, 1, 2, 9
2. Letters: a, b, c, d, ..., z, A, B, ... Z
3. Characters: ESC, DEL, Pg Up
4. Symbols: @, \$, %, etc

ASCII Code

ASCII - American Standard Code for Info Interchange

ASCII is the most common character encoding for English text

ASCII code is used for numerical representation of a character (e.g. A, B, C, a, b, c, etc) in computer system.

ASCII codes represent text in computers, communication equipment, and other devices that use text.

Most modern character-encoding schemes are based on ASCII

ASCII code does not include characters from other languages or symbols used in **mathematics**, **science**, or other fields.

ASCII Code

Code	Usage
0-31	Unprintable control codes and are used to control peripherals such as printers.
32-127	Printable characters and represent letters, digits, punctuation marks, and a few miscellaneous symbols (e.g Code 127 = DEL key).
128-255	Contain the Microsoft Windows Latin-1 extended characters.

ASCII Code (Unprintable control codes)

DEC	OCT	HEX	BIN	Symbol	Description
0	0	0	0	NUL	Null char
1	1	1	1	SOH	Start of Heading
2	2	2	10	STX	Start of Text
3	3	3	11	ETX	End of Text
4	4	4	100	EOT	End of Transmission
5	5	5	101	ENQ	Enquiry
6	6	6	110	ACK	Acknowledgment
7	7	7	111	BEL	Bell
8	10	8	1000	BS	Back Space
9	11	9	1001	HT	Horizontal Tab
10	12	0A	1010	LF	Line Feed
11	13	0B	1011	VT	Vertical Tab
12	14	0C	1100	FF	Form Feed
13	15	0D	1101	CR	Carriage Return
14	16	0E	1110	SO	Shift Out / X-On
15	17	0F	1111	SI	Shift In / X-Off

ASCII Code (Unprintable control codes)

DEC	OCT	HEX	BIN	Symbol	Description
16	20	10	10000	DLE	Data Line Escape
17	21	11	10001	DC1	Device Control 1 (oft. XON)
18	22	12	10010	DC2	Device Control 2
19	23	13	10011	DC3	Device Control 3 (oft. XOFF)
20	24	14	10100	DC4	Device Control 4
21	25	15	10101	NAK	Negative Acknowledgement
22	26	16	10110	SYN	Synchronous Idle
23	27	17	10111	ETB	End of Transmit Block
24	30	18	11000	CAN	Cancel
25	31	19	11001	EM	End of Medium
26	32	1A	11010	SUB	Substitute
27	33	1B	11011	ESC	Escape
28	34	1C	11100	FS	File Separator
29	35	1D	11101	GS	Group Separator

<http://www.ascii-code.com/>

ASCII Code (Printable characters)

DEC	OCT	HEX	BIN	Symbol	Description
30	36	1E	11110	RS	Record Separator
31	37	1F	11111	US	Unit Separator
32	40	20	100000		Space
33	41	21	100001	!	!
34	42	22	100010	"	"
35	43	23	100011	#	#
36	44	24	100100	\$	\$
37	45	25	100101	%	%
38	46	26	100110	&	&
39	47	27	100111	'	'
40	50	28	101000	(	(
41	51	29	101001	)	)
42	52	2A	101010	*	*
43	53	2B	101011	+	+
44	54	2C	101100	,	,
45	55	2D	101101	-	-
46	56	2E	101110	.	.

<http://www.ascii-code.com/>

ASCII Code (Printable characters)

DEC	OCT	HEX	BIN	Symbol	Description
47	57	2F	101111	/	/
48-57	60-71	30-39	110000- 111001	0-9	0-9
59	73	3B	111011	;	;
60	74	3C	111100	<	<
61	75	3D	111101	=	=
62	76	3E	111110	>	>
63	77	3F	111111	?	?
64	100	40	1000000	@	@
65-90	101-132	41-5A	1000001- 1011010	A-Z	A-Z
91	133	5B	1011011	[	[
92	134	5C	1011100	\	\
93	135	5D	1011101	]	]
94	136	5E	1011110	^	^

<http://www.ascii-code.com/>

ASCII Code (Printable characters)

DEC	OCT	HEX	BIN	Symbol	Description
95	137	5F	1011111	`	`
96	140	60	1100000	~	~
97-122	141-172	61-7A	1100001- 1111010	a-z	a-z
123	173	7B	1111011	{	{
124	174	7C	1111100		
125	175	7D	1111101	}	}
126	176	7E	1111110	~	~
127	177	7F	1111111		Delete

<http://www.ascii-code.com/>

ASCII Code

Dec	Hx	Oct	Chr	Dec	Hx	Oct	Html	Chr	Dec	Hx	Oct	Html	Chr	Dec	Hx	Oct	Html	Chr
0	0	000	NUL (null)	32	20	040	 	Space	64	40	100	@	@	96	60	140	`	`
1	1	001	SOH (start of heading)	33	21	041	!	!	65	41	101	A	A	97	61	141	a	a
2	2	002	STX (start of text)	34	22	042	"	"	66	42	102	B	B	98	62	142	b	b
3	3	003	ETX (end of text)	35	23	043	#	#	67	43	103	C	C	99	63	143	c	c
4	4	004	END (end of transmission)	36	24	044	$	\$	68	44	104	D	D	100	64	144	d	d
5	5	005	ENQ (enquiry)	37	25	045	%	%	69	45	105	E	E	101	65	145	e	e
6	6	006	ACK (acknowledge)	38	26	046	&	&	70	46	106	F	F	102	66	146	f	f
7	7	007	BEL (bell)	39	27	047	'	'	71	47	107	G	G	103	67	147	g	g
8	8	010	BS (backspace)	40	28	050	(	(	72	48	110	H	H	104	68	150	h	h
9	9	011	TAB (horizontal tab)	41	29	051	)	)	73	49	111	I	I	105	69	151	i	i
10	A	012	LF (NL line feed, new line)	42	2A	052	*	*	74	4A	112	J	J	106	6A	152	j	j
11	B	013	VT (vertical tab)	43	2B	053	+	+	75	4B	113	K	K	107	6B	153	k	k
12	C	014	FF (NP form feed, new page)	44	2C	054	,	,	76	4C	114	L	L	108	6C	154	l	l
13	D	015	CR (carriage return)	45	2D	055	-	-	77	4D	115	M	M	109	6D	155	m	m
14	E	016	SO (shift out)	46	2E	056	.	.	78	4E	116	N	N	110	6E	156	n	n
15	F	017	SI (shift in)	47	2F	057	/	/	79	4F	117	O	O	111	6F	157	o	o
16	10	020	DLE (data link escape)	48	30	060	0	0	80	50	120	P	P	112	70	160	p	p
17	11	021	DC1 (device control 1)	49	31	061	1	1	81	51	121	Q	Q	113	71	161	q	q
18	12	022	DC2 (device control 2)	50	32	062	2	2	82	52	122	R	R	114	72	162	r	r
19	13	023	DC3 (device control 3)	51	33	063	3	3	83	53	123	S	S	115	73	163	s	s
20	14	024	DC4 (device control 4)	52	34	064	4	4	84	54	124	T	T	116	74	164	t	t
21	15	025	NAK (negative acknowledge)	53	35	065	5	5	85	55	125	U	U	117	75	165	u	u
22	16	026	SYN (synchronous idle)	54	36	066	6	6	86	56	126	V	V	118	76	166	v	v
23	17	027	ETB (end of trans. block)	55	37	067	7	7	87	57	127	W	W	119	77	167	w	w
24	18	030	CAN (cancel)	56	38	070	8	8	88	58	130	X	X	120	78	170	x	x
25	19	031	EM (end of medium)	57	39	071	9	9	89	59	131	Y	Y	121	79	171	y	y
26	1A	032	SUB (substitute)	58	3A	072	:	:	90	5A	132	Z	Z	122	7A	172	z	z
27	1B	033	ESC (escape)	59	3B	073	;	;	91	5B	133	[	[	123	7B	173	{	{
28	1C	034	FS (file separator)	60	3C	074	<	<	92	5C	134	\	\	124	7C	174	|	
29	1D	035	GS (group separator)	61	3D	075	=	=	93	5D	135	]	]	125	7D	175	}	}
30	1E	036	RS (record separator)	62	3E	076	>	>	94	5E	136	^	^	126	7E	176	~	~
31	1F	037	US (unit separator)	63	3F	077	?	?	95	5F	137	_	_	127	7F	177		DEL

Source: www.LookupTables.com

Extended ASCII characters

128	Ç	160	á	192	Ł	224	ó
129	ü	161	í	193	ł	225	ô
130	ë	162	ó	194	Ł	226	õ
131	â	163	ú	195	ł	227	ö
132	ä	164	ñ	196	Ł	228	ó
133	à	165	Ñ	197	ł	229	ô
134	â	166	•	198	Ł	230	õ
135	ç	167	°	199	ł	231	ö
136	ê	168	€	200	Ł	232	÷
137	ë	169	®	201	ł	233	ü
138	è	170	™	202	Ł	234	ý
139	ï	171	½	203	ł	235	ÿ
140	í	172	¼	204	Ł	236	ÿ
141	ì	173	»	205	ł	237	ÿ
142	Ä	174	«	206	Ł	238	ÿ
143	Å	175	»	207	ł	239	ÿ
144	É	176	•	208	Ł	240	ÿ
145	æ	177	•	209	ł	241	ÿ
146	Æ	178	•	210	Ł	242	ÿ
147	ö	179	•	211	ł	243	ÿ
148	ø	180	•	212	Ł	244	ÿ
149	ò	181	•	213	ł	245	ÿ
150	ó	182	•	214	Ł	246	ÿ
151	ù	183	•	215	ł	247	ÿ
152	ý	184	•	216	Ł	248	ÿ
153	Û	185	•	217	ł	249	ÿ
154	Ü	186	•	218	Ł	250	ÿ
155	ø	187	•	219	ł	251	ÿ
156	£	188	•	220	Ł	252	ÿ
157	Ø	189	•	221	ł	253	ÿ
158	×	190	•	222	Ł	254	ÿ
159	ƒ	191	•	223	ł	255	nbsp

<https://theasciicode.com.ar/>

frequently-used (spanish language)	vowels acute accent (spanish language)	vowels with diaresis
ñ alt + 164	á alt + 160	ä alt + 132
Ñ alt + 165	é alt + 130	ë alt + 137
@ alt + 64	í alt + 161	ï alt + 139
¿ alt + 168	ó alt + 162	ö alt + 148
? alt + 63	ú alt + 163	ü alt + 129
¡ alt + 173	Á alt + 181	Ä alt + 142
! alt + 33	É alt + 144	Ë alt + 211
: alt + 58	Í alt + 214	Ï alt + 216
/ alt + 47	Ó alt + 224	Ö alt + 153
\ alt + 92	Ú alt + 233	Ü alt + 154

mathematical symbols	commercial / trade symbols	quotes and parenthesis
½ alt + 171	\$ alt + 36	" alt + 34
¼ alt + 172	£ alt + 156	' alt + 39
¾ alt + 243	¥ alt + 190	(alt + 40
¹ alt + 251	¢ alt + 189	) alt + 41
³ alt + 252	¤ alt + 207	[alt + 91
² alt + 253	® alt + 169	] alt + 93
f alt + 159	© alt + 184	{ alt + 123
± alt + 241	ª alt + 166	} alt + 125
× alt + 158	º alt + 167	« alt + 174
÷ alt + 246	° alt + 248	» alt + 175

ASCII Code

Hi hello World

Char	Decimal	Binary
H	72	01001000
i	105	01101001
Space	32	0100000
h	104	01101000
e	101	01100101
l	108	01101100
l	108	01101100
o	111	01101111
Space	32	0100000
W	87	01010111
o	111	01101111
r	114	01110010
l	108	01101100
d	100	01100100

How does a computer process data?



Hi hello World



Hi hello World

Char	Decimal	Binary
H	72	01001000
i	105	01101001
Space	32	0100000
h	104	01101000
e	101	01100101
l	108	01101100
l	108	01101100
o	111	01101111
Space	32	0100000
W	87	01010111
o	111	01101111
r	114	01110010
l	108	01101100
d	100	01100100

Unicode Transformation Format (UTF)

- UTF-8 (8 bits) - is a character encoding scheme that was developed to address the need for a universal character encoding capable of representing text from all the world's writing systems → i18n & multilingual applications.
- it was designed by Ken Thompson, Rob Pike, and others at Bell Labs in 1992.
- It extends beyond ASCII to include a vast range of characters from different scripts, languages, and symbols
- It introduced variable-length encoding to represent characters using one, two, three, or even four bytes (8, 16, 24, or 32 bits).

我去北京 我去北京
我去北京 我去北京
我去北京 我去北京

모든 인간은 태어날 때부터
자유로우며 그 존엄과 권리에
있어 동등하다. 인간은 천부적으로
이성과 양심을 부여받았으며 서로
형제애의 정신으로 행동하여야 한다.

ああ、天の河と、島村も振り仰いだとたん、天の
河のなかへ船がふりと浮き上つてゆくやうだつた。天
の河の明るさが島村を照らして近かつた。旅の
芭蕉が荒海の上に見たのは、このやうにあざやかな天
の河の大きであつたか。樺の天の河は後の大地を素
肌で巻かうとして、直ぐそこに降りて來てゐる。空ろ
しい舞めかしさだ。島村は自分の小さい影が地上から
逆に天の河へ寫つてゐるに感じた。天の河にいつは
いの星が一つ一つ見えるばかりでなく、ところどころ
光雲の銀砂子も一粒一粒見えるほど濃み濃り、しかも
天の河の底なしの深さが視網を吸ひ込んで行つた。

Unicode Transformation Format (UTF)

- UTF-8 is widely used on the internet, in software development, and in operating systems.
- UTF-8 was designed to be backward-compatible with ASCII, ASCII has become a subset of UTF-8
- UTF-8 has become the de facto standard for character encoding in modern computing due to its versatility, backward compatibility with ASCII, and widespread support.
- The latest version of Unicode contains a collection of 120,000+ characters covering 129 modern and historic scripts, as well as multiple symbol sets.

我去北京 我去北京
我去北京 我去北京
我去北京 我去北京

모든 인간은 태어날 때부터
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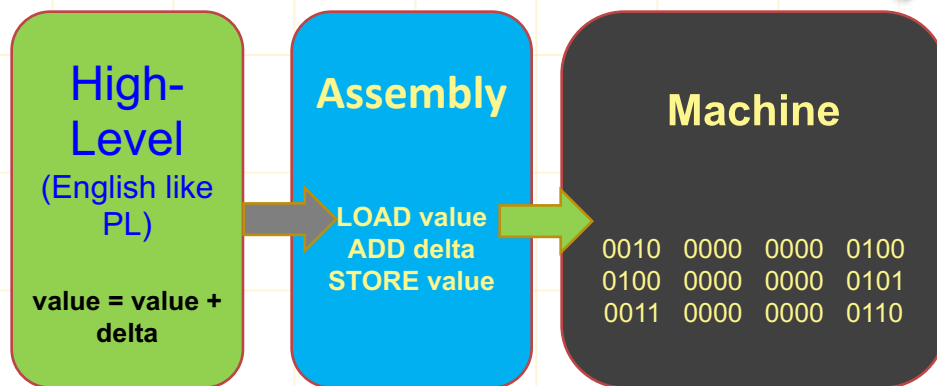
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天の河の底なしの深さが視網を吸ひ込んで行つた。

Other modern Character encoding schemes

- UTF-16 -- is another Unicode encoding that uses 16-bit code units to represent characters.
 - It can represent a broader range of characters compared to UTF-8
 - Commonly used in certain applications and platforms, including Windows.
- UTF-32 -- uses a fixed 32-bit encoding for each character.
 - It provides a direct and fixed-width representation of Unicode characters
 - less space-efficient than UTF-8 and UTF-16.
- Windows 1252 -- an extension of ISO 8859-1 and is commonly used in Windows environments for Western European languages.
- Shift JIS -- Used primarily in Japanese computing environments, Shift JIS is a variable-length encoding for Japanese characters.

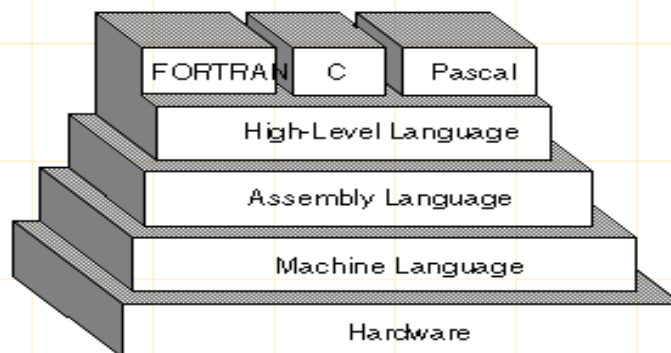
character	encoding	bits
A	UTF-8	01000001
A	UTF-16	00000000 01000001
A	UTF-32	00000000 00000000 00000000 01000001
あ	UTF-8	11100011 10000001 10000010
あ	UTF-16	00110000 01000010
あ	UTF-32	00000000 00000000 00110000 01000010

Programming Languages



Since machine language is hard to understand, programmers commonly use more English-like or high-level languages such as C/C++, Java, etc to write programs which are then translated into machine language (called a low level language) by an assembler, compiler or interpreter.

Programming Languages



Since machine language is hard to understand by humans, programmers commonly use more English-like or high-level languages such as C/C++, Java, etc to write programs which are then translated into machine language (called a low level language) by an assembler, compiler or interpreter.

Source: <http://www.webopedia.com/>

```
#include <stdio.h>

int
main(int argc, char * argv[])
{
    printf("hello, world\n");
}
```



```
10000011 00000001 00010001 00000000 00111101 11111100 01110100 00111101
00000000 01000000 00000000 00000000 00000000 00000000 00000000 00000000
10010000 00000000 00000000 00000000 01010000 00000000 00000111 00110000
00001011 00000001 00001011 00000011 00001010 00000000 00000000 00000000
00000000 00100000 00000000 00000000 00000000 00000000 00000000 00000000
00000000 00100000 00000000 00000000 00000000 00000000 00000000 00000000
00000000 00000000 00000000 00000000 00000000 00000000 00000000 00000000
01110000 00010000 00000000 00100000 00000001 00000000 00000000 00000000
00000000 00000000 00000000 00100000 00000001 00000000 00000000 00000000
00000000 00000000 00000000 01000000 00000001 00000000 00000000 00000000
00000000 00100000 00000000 01000000 00000001 00000000 00000000 00000000
11111111 11111111 11111111 11111111 11111111 11111111 11111111 11111111
10010000 10000000 00000000 01000000 00000001 00000000 00000000 00000000
00101110 01100100 01111001 01101110 01100001 01101101 01101001 01100011
10110000 00000100 00000000 00100000 00000001 00000000 00000000 00000000
10110000 00000100 00000000 00100000 00000001 00000000 00000000 00000000
10100000 00000001 00000000 00000000 00000000 00000000 00000000 00000000
10110000 00000100 00000000 00000000 00000000 00000000 00000000 00000000
00000000 00000000 00000000 00000000 00000000 00000000 00000000 00000000
00000000 00000000 00000000 00000000 00000000 00000000 00000000 00000000
00000000 00000000 00000000 00000000 00100000 00000000 00000000
```

[...]

Representing information as Bits

- Information comes in many forms
 - Words, numbers, pictures
 - Sound, movies
- Bits as Numbers - Binary number system
- Bits as Symbols - ASCII, Unicode
- Bits as Instructions in Programs - Programs stored as collection of bits

1. Byte = 8 bits
2. Kilobyte (KB) = 1024 bytes
3. Megabytes (MB) = 1 million bytes
4. Gigabytes (GB) = 1 billion bytes
5. Terabytes (TB) = 1 trillion bytes



Thank you!